INSTACART

Instacart Project Scope

Version 1.2
Verified CI and BigQuery reconciliation

Current status, practical boundaries and evidence-based completion

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Reviewed through 1 November 2026
Committed code baseline: `c90c4f0`

SQL / BI / Python	71 hosted tests passed	5 cloud tables checked
55 result rows matched	dbt compatibility next	Current commit pending

Completed work and planned extensions are clearly separated. Future milestones become complete only after their acceptance evidence is recorded.

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Purpose of this document

This document defines what the project is intended to demonstrate, what has been completed, what remains planned, and what evidence is required before a capability is described as finished. The roadmap explains the implementation order. The detailed reports explain the completed work.

The project began as SQL Server and Power BI analysis of Instacart shopping behavior. It now also includes a verified local Python loading and validation workflow, hosted continuous integration, and a five-table BigQuery subset with selected SQL Server result reconciliation. Later extensions add dbt transformations and a carefully evaluated prediction task. Each extension has a separate acceptance decision; a plan does not establish an implemented skill.

1. Project objective and audience

Objective: build a reproducible portfolio project that explains historical shopping behavior and demonstrates how analytical data can be prepared, checked, recovered, and eventually used for predictive analysis.

Audience	What the project should make clear
Technical reviewer	Data grain, SQL logic, contracts, dependency order, failure handling, tests, and implementation limits
Hiring professional	What Saurabh implemented, the scale handled, the questions answered, and the evidence available
Student or practitioner	How to reproduce the work and understand each stage without needing undocumented steps
Project owner	What is complete, what comes next, and which deliverables belong to each milestone

The portfolio demonstrates project experience. It does not claim years of production data engineering, enterprise GCP administration, or on-call support from a local demonstration.

1.1 Business questions already answered

The completed analysis asks which products are most common, how widely they occur across orders, how often entries are marked reordered, which departments lead volume or reorder rate, when ordering peaks, and how basket sizes vary.

Verified finding	Interpretation
Banana appears in 472,565 prior orders; 14.70% penetration	Reach across historical orders
19,126,536 reordered entries; 58.97% reorder rate	Share of entries marked as previously purchased
Produce contributes 9,479,291 entries	Department entry volume
Dairy eggs has a 67.00% reorder rate	Department repeat-marked entry share
Hour 10 has 271,885 orders	Peak order hour in the analyzed baseline
Mean basket 10.09; mode 5; range 1-145	Distinct products per prior order

These are descriptive historical results. Product entries are not item quantities or revenue. Reorder rate is not customer retention. There is no price, profit, transaction date, or external economic data in the declared analytical scope.

2. Dataset boundaries and metric definitions

2.1 Source inputs

The source directory on the development machine is `D:\DataProjects\data-download`. Files remain outside the Git repository. Download/access conditions follow the source dataset's terms.

Input	Rows or role	Current use
`departments.csv`	21 rows	SQL baseline and Python automation
`aisles.csv`	134 rows	SQL baseline and Python automation
`products.csv`	49,688 rows	SQL baseline and Python automation
`orders.csv`	3,421,083 rows	Metadata for prior, train, and test cohorts
`order_products__prior.csv`	32,434,489 rows	Historical order-product entries
`order_products__train.csv`	Available locally; product outcomes for train orders	Future ML labels; outside current five-file baseline

The five automated inputs have a manifest and recorded byte sizes/headers/SHA-256 values. This is a fixed reference dataset, not a live daily feed. Changing source bytes requires an explicit version/contract decision.

2.2 Cohorts and grain

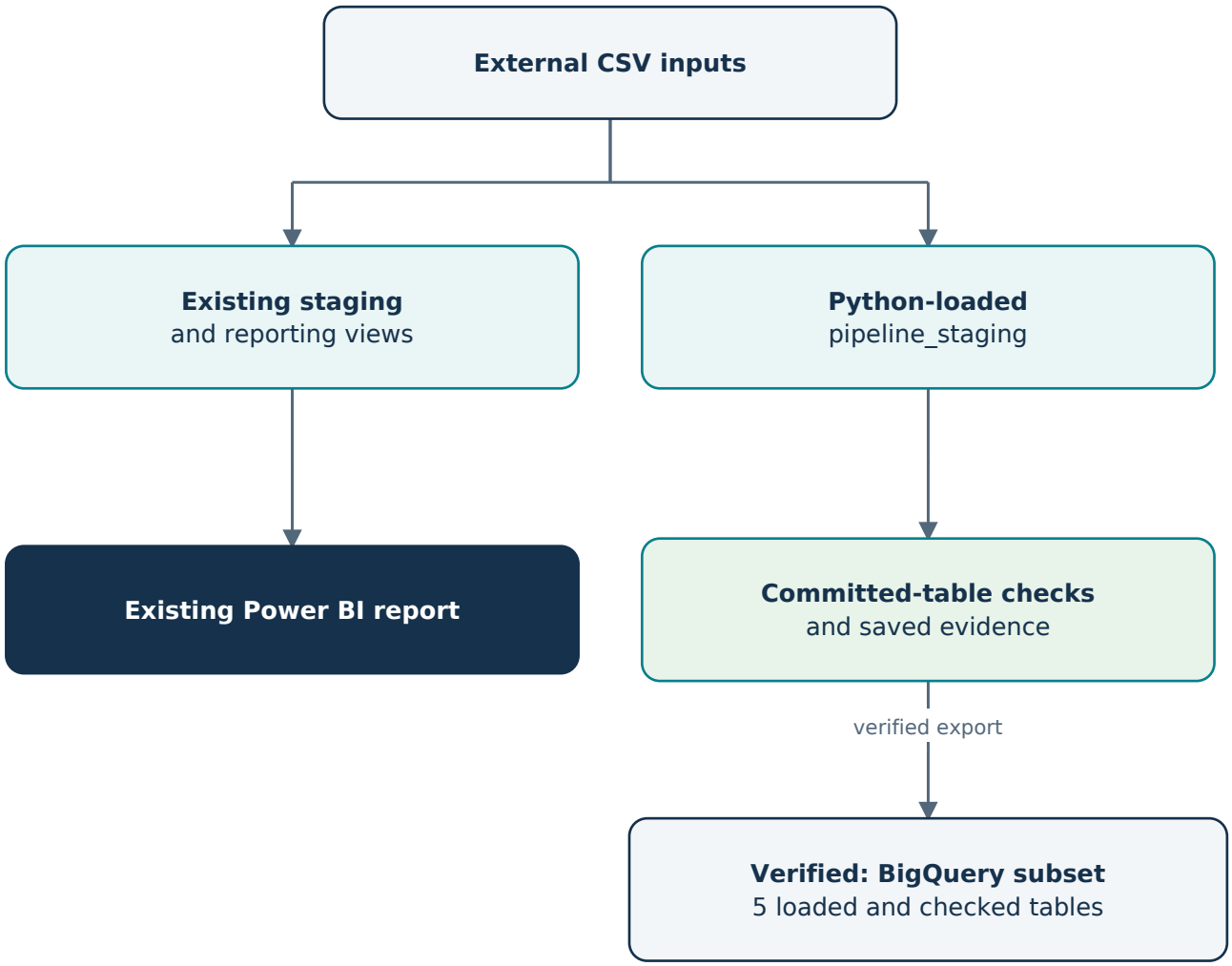
Object or cohort	Definition
Prior orders	3,214,874 historical orders used by the descriptive reporting views
Train orders	131,209 order metadata rows; associated product-label CSV is separate
Test orders	75,000 order metadata rows; final product answers are not available in the shown inputs
Users	206,209 distinct users in all-order metadata
Product grain	One row per product ID
Order grain	One row per order ID
Entry grain	One row per `(order_id, product_id)` pair

The prior/train/test labels describe dataset roles. They are not an automatically valid modeling split. A future model must define its prediction target, history cutoff, and evaluation partitions explicitly.

2.3 Agreed metric meanings

Metric	Definition and boundary
Order penetration	Orders containing a product divided by all prior orders in the intended order/day context
Reorder rate	Reordered entry rows divided by all entry rows in the selected context
Basket size	Distinct product count in an order, not summed quantities
Order gap	Days-since-prior-order field; first-order NULL is meaningful; 0 is valid; values are capped at 30
Day code	Dataset values 0-6; no assumed weekday names
Product-filter behavior	Products filter OrderProducts; they do not filter back into Orders in the established single-direction model

3. Current architecture and ownership boundaries



Current data paths and planned extensions

Component	Responsibility	Status
Original `staging`	Imported baseline for established SQL work	Implemented
Existing `reporting` views	Product dimension, prior orders, prior entries	Implemented
Power BI Desktop	Three-page local descriptive report	Implemented locally
`pipeline_staging`	Five permanent tables populated by Python loaders	Implemented and verified
Temporary SQL tables	Validate input before target replacement	Implemented
Local JSON reports	File, loader, check, and coordinator execution history	Implemented

Component	Responsibility	Status
BigQuery `instacart_raw`	Five loaded subset tables; manual quality checks and three analytical comparisons	Verified in the owner session; SQL/evidence commit pending
Future dbt/ML outputs	Separately named transformation and prediction outputs	Planned or gated

The new Python workflow does not rebuild reporting views from `pipeline_staging` or refresh the Power BI report. Connecting those paths is a future integration decision with its own reconciliation checks. Current analysis remains reproducible through the established SQL/reporting path.

4. Completed deliverables and acceptance evidence

4.1 SQL and Power BI

Completed assets include numbered SQL scripts 01-21 plus auxiliary script 00, five original imports, quality checks, business analyses, three reporting views, seven DAX measures, and three final local report pages: Overview, Products & Reordering, and Ordering & Baskets.

Day Code is synced across all three pages. Department filtering stays local to Products & Reordering. SQL and report metric definitions were reconciled at the analytical milestone. The technical SQL report contains the per-script findings and DAX details.

The PBIX is LFS-tracked. An earlier pointer recorded 208,487,698 bytes. A fresh clone/download-and-open test on the delivered binary remains unverified in this record. Local completion and clean delivery verification are separate acceptance checks.

4.2 Python loading and validation

Capability	Verified result
Reporting quality checks	18 checks across five groups; saved failure/recovery evidence
File identity enforcement	All five real inputs matched the recorded baseline
Initial database loading	Five real table loads, including 32,434,489 prior entries
Safe replay	All five loaders reported UNCHANGED in the complete coordinator run
Transaction recovery	Department failure after clearing the target rolled back; SSMS verified 21 unique IDs afterward
Final accepted-table checks	Five tables passed 27 named metric checks
Coordinator behavior	Missing inputs stopped execution before any loader; corrected input run passed all seven stages
Automated code checks	Original 53-test pipeline suite; expanded exporter suite now 71 tests, local and hosted PASS at c90c4f0
Preserved evidence	Nine selected JSON reports committed

The successful combined replay took 163.175 seconds in its saved local run. That is one observed replay duration, not an initial-load benchmark or runtime guarantee. Mocks cover controlled code behavior; live SQL

execution supplies the separate loading/rollback evidence.

4.3 Evidence checkpoints

Checkpoint	What it establishes
`76f0b23`	Earlier analytical documentation and GitHub delivery checkpoint
`4e6536a`	Reporting validators, combined reports, and saved evidence
`743c62a`	Input manifest, fingerprint baseline, and 15 input tests
`bfc1763`	Completed local loading/validation workflow and 53-test suite
`dcb145d`	Published Python report, project scope, roadmap, and documentation guide
`c863046`	First hosted CI workflow; 53 tests passed on main
`a411277` / `647d9c2`	Deliberate regression detected, then corrected pass on the demonstration branch
`aa46eeb`	Published CI failure/recovery documentation
`c90c4f0`	Committed verified subset exporter and selected evidence; hosted CI passed 71 tests
Owner session, through 1 November	Five BigQuery loads/quality checks and 55 matching analytical result rows; current source/evidence publication pending

The original Python report and planning documents were published at dcb145d. CI operation and its failure/recovery exercise are separately verified; c90c4f0 passed all 71 expanded tests on hosted Windows. The owner supplied successful BigQuery validation and paired analytical screenshots. The BigQuery runbook and review record preserve the reviewed cloud evidence. Current cloud SQL/docs/evidence await the next GitHub Desktop commit. Earlier reports retain their historical checkpoints and test counts.

4.4 BigQuery subset and reconciliation

The deterministic 5,000-user subset contains 81,832 order metadata rows (76,832 prior, 3,200 train, 1,800 test), 761,750 prior entries, 28,854 referenced products, and full small lookups. The exported files matched their size/hash manifest. Five manual console loads and the configured quality checks passed in BigQuery. All 24 hourly, 21 department, and 10 top-product result rows matched SQL Server, including the displayed two-decimal rates.

The review is based on the owner's paired screenshots and observed terminal output, not a database API export or automated full-row parity checker. The subset is not a random sample. BigQuery uses project `instacart-analytics-practice`, dataset `instacart_raw`, US data location, and us-central1 saved-code location as configured in the guided setup. Schema/load job metadata was not independently exported. dbt, programmatic cloud loading, scheduling, and deployment remain unimplemented.

5. Approved learning direction and future deliverables

This table distinguishes accepted extensions from the remaining learning direction. It does not authorize paid resources. The companion roadmap defines dependencies and checks.

Extension	Intended outcome	Evidence required
Continuous integration	Core milestone complete: hosted push checks, deliberate failure, corrected pass	CI runbook and selected evidence; PR/manual triggers configured but not separately exercised
Cloud SQL	Scoped batch demonstration verified; publication pending	Five table checks and 24/21/10 matching analytical rows; runbook and screenshot review
dbt	Reusable documented SQL models and quality tests	Model dependency graph, test results, generated documentation
Machine learning	Predict previously purchased products in a next order	Leakage-controlled splits, baseline/model metrics, error analysis, reproducible batch predictions
Operational tracking	Record tasks, defects, and resolution evidence	Small board or backlog with acceptance/evidence links
Airflow	Schedule/coordinate already-working compatible tasks	DAG, controlled failure, correct downstream blocking, successful rerun
Continuous delivery	Produce a traceable tested release artifact	Release version, dependency record, smoke check, previous-version recovery
Automatic deployment	Update and verify a chosen running target	Optional; requires an actual target and operating/cost decision
Portfolio communication	Explain implemented work and connect with others	Separate LinkedIn drafting/review discussion when initiated

The first ML task is a reordered-product recommendation/classification task over previously purchased candidates. Forecasting revenue, predicting exact quantities, discovering all brand-new products, and real-time serving are separate questions. Model improvement over the baseline will be measured rather than promised.

6. Technologies: retain, add, or defer

Technology	Scope choice	Why
SQL Server, SSMS, Power BI	Retain established analytics	Preserve validated logic, grains, and report behavior
Python 3.14 + uv + pyodbc	Retain local workflow	Existing reproducible environment and successful loading evidence
`unittest`	Retained for hosted CI	Expanded 71-test suite passed at c90c4f0; framework migration unnecessary
GitHub Actions	Implemented core CI	Hosted Windows tests detected the regression and verified the correction
BigQuery sandbox	Verified small cloud branch	Five loaded/checked tables and selected aggregate reconciliation with SQL Server
dbt + compatible adapter	After cloud/subset proof	Model dependencies, SQL tests, and documentation
SQL aggregates + scikit-learn	First ML path	Reduce large history to manageable features; begin with interpretable baselines
DuckDB	Conditional fallback	Local dbt practice if cloud constraints block the intended branch; label the outcome accurately
Airflow via WSL2/Linux environment	Later, gated by compatibility	Orchestrate working jobs rather than fix their logic inside a scheduler
Docker / image registry	Conditional delivery format	Useful only after proving a runnable environment and database connection path
Jira	Lightweight tracking option	Demonstrate clear task/defect ownership without making a board the project bottleneck

Future libraries should be pinned only after compatibility is checked. Airflow/dbt may use a separate environment. The successful Windows integrated-authentication connection is not automatically portable into a Linux container or hosted runner.

7. Explicit exclusions and constraints

Boundary	Scope rule
Production experience	Local project runs do not establish enterprise availability or on-call operations
Unattended operation	Current pipeline is manually started; scheduling/retries/alerts remain planned
Full-pipeline atomicity	Each loader commits separately; later failure can leave earlier successful commits
Concurrency	Run one local pipeline at a time; there is no global run lock
Snapshot consistency	Final table checks are separate queries; avoid concurrent edits/loads
Data changes	Baseline and fixed counts target selected static data; changed data needs a reviewed contract
Schema evolution	Targets are created if missing; schema migrations are not implemented
Source authenticity	Local fingerprints identify selected bytes; they are not independently published authenticity proofs
BI delivery	Power BI Service publication and scheduled refresh are outside current completion
Resource use	Approximately 12 GB RAM; shared SQL/Power BI/container workloads require capacity checks
Costs	No paid cloud resource, billing upgrade, or automatic deployment is established by this plan
Advanced platforms	Kubernetes, streaming, Spark clusters, broad enterprise GCP administration, and deep learning are outside the initial roadmap

For BigQuery, the sandbox restricts capabilities including DML and streaming and expires stored tables/views after 60 days. Keep a reproducible small batch demonstration and recheck the current quotas before loading [R1]. For Airflow's Docker quick start, the documentation calls for at least 4 GB of container memory and recommends more; the laptop needs a feasibility check before that stage [R2].

8. Definition of done and evidence policy

A milestone is done when its implementation exists, its stated acceptance checks pass, representative evidence is saved, the operating instructions explain how to reproduce it, and its limitations are recorded. Documentation creation, local review, Git commit, and remote publication should be marked separately when they occur at different times.

Evidence kind	Storage policy
Source code/configuration/tests	Commit to Git with a meaningful milestone message
Raw CSV data and local environment	Keep external/recreated; follow dataset terms and the existing ignore policy
Routine JSON run history	Keep under ignored `logs` folders
Selected pass/failure/recovery reports	Copy into `docs/validation_evidence` with descriptive names
Cloud and ML demonstrations	Store small manifests, metric summaries, reproducible scripts, and permitted sample outputs
Models/images/releases	Choose appropriate artifact/release storage; do not add large generated binaries blindly to ordinary Git
Documentation diagrams	Keep local relative links and editable diagram sources with the documents

No accuracy threshold, test-count growth target, deadline, or job outcome is promised. Future test counts can change; passing the relevant suite matters more than retaining the number 53 forever.

9. Scope-change process

Before adding a substantial extension, record its question, proposed output, affected data path, technology, resource/cost impact, and acceptance evidence. Mark it Planned or Conditional. Start it after its prerequisites are accepted. Update the scope/roadmap after completion rather than presenting planned work as already implemented.

Decision	Current resolution
Move established Power BI source to new automation tables?	Open; preserve current views until a separately reconciled integration is chosen
Replace unittest with pytest?	No current need; existing suite is already used by CI
Use a different Python version for a new tool?	Allowed in an isolated environment after compatibility review
Run full database imports on each hosted CI push?	Outside the first CI scope; use deterministic offline tests
Choose a paid cloud deployment now?	Deferred; no target selected
Draft another LinkedIn post now?	Deferred to the separate discussion requested by Saurabh

10. Immediate handoff and maintenance

1. Review the BigQuery runbook, screenshot evidence and updated scope/roadmap.
2. Save these files under `docs`, preserving diagrams and screenshots beside their relative links.
3. Commit/push the eight cloud SQL files, three local reference queries and current documentation/evidence through GitHub Desktop; verify the resulting hosted CI run.
4. Resolve the outstanding PBIX delivery smoke check when convenient; it is separate from the cloud result comparisons.
5. Begin DBT-01 with a small adapter/runtime/warehouse compatibility proof before expanding transformations.

Update this scope when accepted deliverables or exclusions change. Update the roadmap after every accepted milestone. Link new evidence to an exact commit or saved report. These documents record planning and current status; they do not start new jobs or schedule future actions.

References

- R1: Google Cloud - BigQuery sandbox
- R2: Apache Airflow - Docker learning setup
- Verified implementation commit
- SQL technical report at the implementation checkpoint